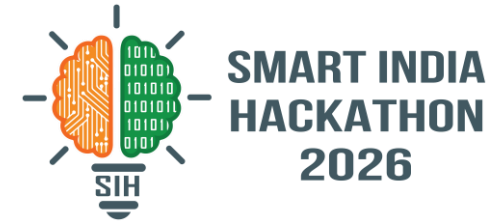
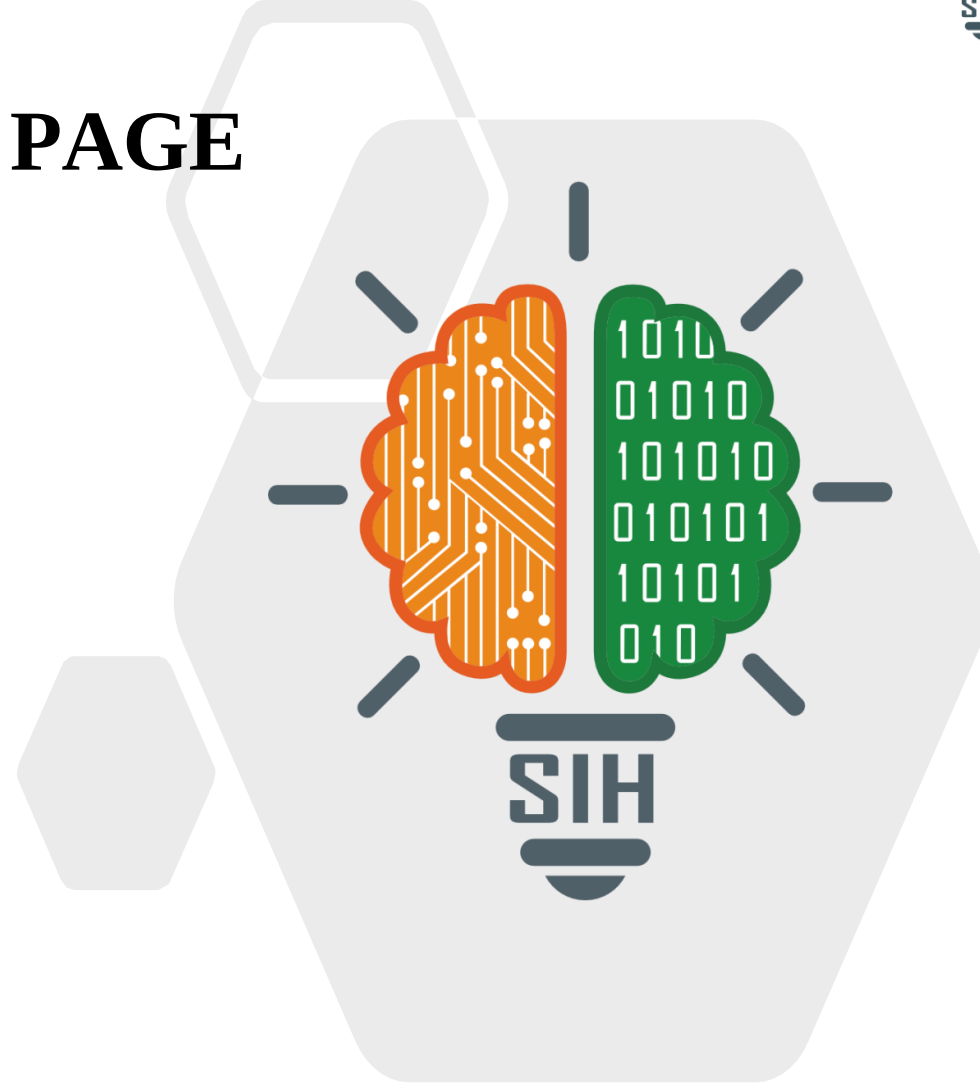


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26106
- **Problem Statement Title** – AI-Powered Email Threat Detection, Geo-Location and Forensic Intelligence Platform
- **Theme** – Blockchain & Cybersecurity
- **PS Category** – Software
- **Team ID** – 128986
- **Team Name** – Cache-Me-Maybe



PRAHARI: AI-Powered Email Threat Detection & Forensic Intelligence Platform

PROBLEM

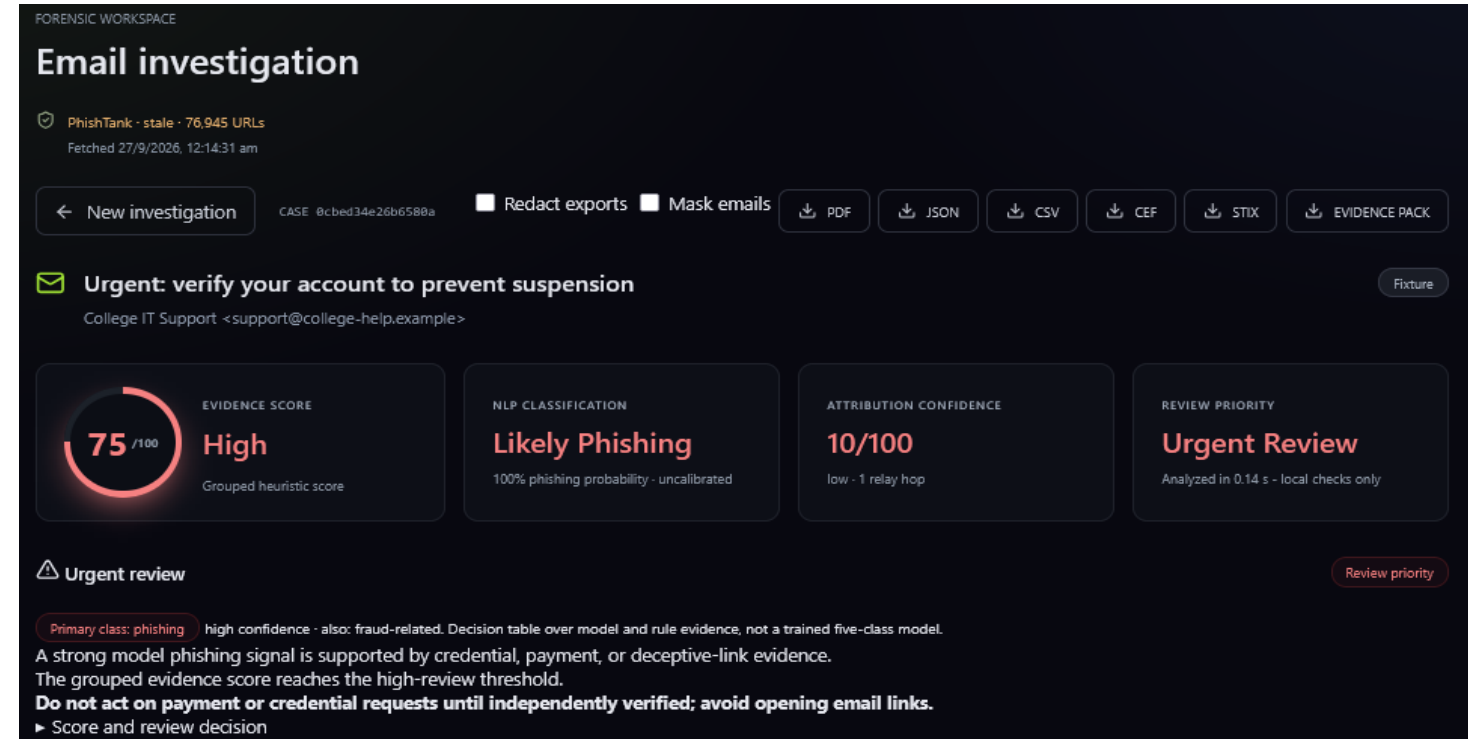
Bank of Baroda (July 2026): An employee email account was compromised, with unauthorized access to certain data. CERT-In handled 29,44,248 cyber incidents in 2025. Modern attacks use AI-written content, impersonation and malicious links, so a single-email filter is not enough.

SOLUTION

- AI-Powered Phishing & BEC Detection
- Sender Authentication (SPF, DKIM, DMARC, ARC)
- Infrastructure Intelligence & Geo-Mapping
- Campaign Correlation & Attribution Confidence
- Tamper-Evident Evidence Preservation

UNIQUENESS

- AI-manipulation detection (prompt injection)
- Three separate outputs: score, confidence, priority
- Attribution confidence, not attacker identity
- Tamper-evident chain (SHA-256 + Bitcoin timestamp)
- Investigator leads and SOC-ready exports



Evidence score	Attribution confidence	Review priority	Class and next step	SOC-ready exports
How concerning the signals are: a heuristic, not a probability.	How far to trust the infrastructure trace.	What the analyst handles first, separate from the score.	One of five classes, with plain-language guidance.	PDF, JSON, CSV, CEF, STIX and evidence pack.

- Screenshot taken from the live, working prototype (built-in sample email)

**ONE EMAIL IN → COMPLETE THREAT
INTELLIGENCE CASE OUT**

One engine behind every entry point: upload, paste, Gmail, browser extension and SMTP gateway. Every flag shows its reason, and missing evidence is never shown as a pass.

808 automated tests · ~75 ms per email · no paid APIs needed



7

Outcome

- Detect threats
- Investigate attacks
- Attribute infrastructure
- Preserve evidence
- Support SOC operations

6

Output & Integration

- Analyst dashboard
- Forensic reports (PDF, JSON)
- SOC exports (CEF, Splunk, STIX)
- DPDP-aware masking

5

Investigation & Evidence Engine

- Evidence score
- Attribution confidence
- Campaign correlation
- SHA-256 hash chain
- Bitcoin timestamp

4

Infrastructure Intelligence

- ASN and IP geolocation
- Known-bad IP feeds
- Tor exit-node matching
- VPN / proxy hints (optional)

3

Email Security & Authentication

- SPF check
- DKIM check
- DMARC check
- ARC chain check

2

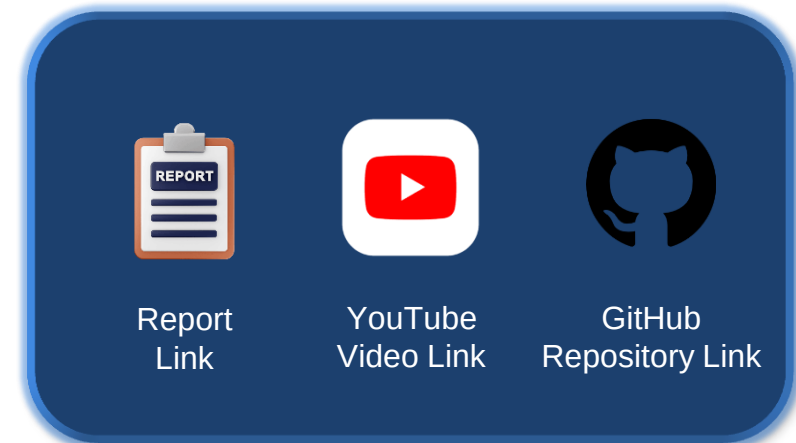
AI & Threat Detection

- Fine-tuned BERT model
- Spoofing and brand-lookalike detection
- BEC and fraud cues
- AI-manipulation detection

1

Development Stack

- React + Vite frontend
- Python FastAPI backend
- Chrome Gmail extension





FEASIBILITY



- **Proven Technologies:** FastAPI, React, BERT, SPF/DKIM/DMARC, VirusTotal, AbuseIPDB, OpenTimestamps.
- **Working Prototype:** detects phishing, BEC and spoofing, and produces forensic cases. 808 automated tests pass.
- **Integration:** SIEM export (Splunk HEC delivery verified) and Gmail real-time monitoring.



CHALLENGES



- **Advanced Threat Evasion:** VPNs, Tor and compromised domains hide the origin.
- **New and Look-Alike Phishing:** unseen campaigns and promotional mail can confuse a text-only model.
- **Data Correlation Complexity:** combining authentication, infrastructure, content and evidence is computationally intensive.



MITIGATION



- **Infrastructure Attribution:** correlate IPs, ASNs and relay paths, with a confidence score.
- **Multi-Layer Detection:** BERT plus authentication, URL and infrastructure evidence, with analyst review.
- **Tamper-Evident Evidence:** SHA-256 hash chain with a Bitcoin timestamp.
- **Continuous Updates:** reputation feeds update without changing the core design.

Tamper-evident by design: every case event is SHA-256 hash-chained to the one before it, and the chain is anchored to Bitcoin with OpenTimestamps



✗ **If any record is edited:** its hash changes, every later link breaks, and verification fails, so tampering is visible (tamper-evident, not tamper-proof).

SECURITY IMPACT



Enhanced Email Security

Detects phishing, BEC, spoofing and impersonation.



Faster Threat Investigation

Turns an email into a full forensic case in seconds.



Infrastructure Intelligence

Reveals relay infrastructure, geolocation and ASN.



Reduced Organizational Risk

Helps reduce fraud, account compromise and reputational damage.

OPERATIONAL BENEFITS



Analyst Efficiency

Automates triage, attribution and evidence collection.



Explainable Decisions

Confidence scores and transparent reasoning for every flag.



Tamper-Evident Evidence

SHA-256 hash chain plus Bitcoin timestamp.



Privacy-Conscious Processing

Local processing; Aadhaar, PAN and UPI masked in exports.

WHO BENEFITS

- **SOC analysts:** faster triage, evidence in one place
- **Cybercrime units:** structured forensic package
- **Banks & PSUs:** early phishing / BEC detection
- **Email admins:** authentication & spoofing visibility
- **Compliance teams:** masked, exportable evidence

VERIFIED IN THE PROTOTYPE

- About 75 ms median analysis time (local)
- About 2 to 4 s with live enrichment
- 808 automated backend tests passing
- Live-checked: Gmail push, Splunk HEC, Bitcoin timestamp (block 968372)
- Pre-delivery SMTP gateway holds high-risk mail (tested end to end)
- Aadhaar, PAN, UPI and mobile numbers masked in exports

Social: affordable email forensics for institutions with small security budgets.

Economic: reduces exposure to BEC fraud and credential theft.

Environmental: runs on an ordinary CPU, with no GPU or cloud servers needed.

Research Survey

RESEARCH AREA	EXISTING APPROACHES	GAP IDENTIFIED	OUR APPROACH
Email Threat Detection	Rule-based filters & blacklists	Poor adaptation to AI-generated phishing	Fine-tuned BERT + multi-signal evidence
Email Authentication	SPF, DKIM, DMARC validation	Authentication alone cannot determine malicious intent	Authentication + content + infrastructure evidence
Threat Investigation	Manual analysis by security teams	Time-consuming and analyst-dependent	Automated threat investigation workflow
Infrastructure Attribution	IP reputation & WHOIS	Limited infrastructure correlation	ASN, Relay Path, GeoIP & Infrastructure Intelligence
Evidence Integrity	Conventional logging systems	Logs can be altered or disputed	SHA-256 hash chaining & OpenTimestamps (Bitcoin) anchoring

References

[1] LLM-Based Multimodal Feature Extraction and Hierarchical Fusion for Phishing Email Detection (2026)

Yuan et al., Electronics 15(2):368. Reports that fusing email text, URL, page and code evidence improves phishing detection.
Available: <https://www.mdpi.com/2079-9292/15/2/368>

[2] Digital Forensics and Phishing Defense: A Literature Review and Gap Analysis (2026)

Laleb, Le & Nguyen, J. Cybersecur. Priv. 6(4):116. Forensic-quality artefacts (headers, DNS, provenance) support attribution and evidential reporting.
Available: <https://doi.org/10.3390/jcp6040116>

[3] PhishLumos: From a Single URL to Campaign-Level Phishing Mitigation (2026)

Chiba, Nakano & Koide, IEEE Access. Shows the value of grouping attacks by shared hosting and registration patterns.
Available: <https://doi.org/10.1109/ACCESS.2026.3696597>

Emerging Trends Incorporated

- Multi-signal phishing detection beyond content
- Campaign-level threat correlation
- Infrastructure attribution intelligence
- Explainable, evidence-backed investigation
- Tamper-evident forensic evidence preservation
- SOC/SIEM-ready security operations